

Erdős Problem #172

EP-172

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Is it true that in any finite colouring of $\mathbb{N}$ there exist arbitrarily large finite A such that all sums and products of distinct elements in A are the same colour?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>